EPPS 6302 Methods of Data Collection and Production: Assignment 5

Government data API

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> What you will produce
>
> A working GovInfo API key, the ten most recent Senate Foreign Relations Committee hearings downloaded two ways, and a short report on how usable the documents are as data.

0. Apply for a GovInfo API key

a. Sign up at https://www.govinfo.gov/api-signup with your name and UTD email. The key is issued by **api.data.gov** and arrives by email; one key works for GovInfo and other federal APIs.
b. Store it in `.Renviron`, not in your code: run `usethis::edit_r_environ()`, add the line `GOVINFO_API_KEY=your-key`, save, and **restart R**.
c. Test it: paste `https://api.govinfo.gov/collections?api_key=YOUR-KEY` into a browser. You should see a JSON list of collections.

> **Treat the key like the Census and YouTube keys.** Never in a `.qmd`, never committed — your website repository is public. `DEMO_KEY` works in place of a key for a first test, but it is tightly rate-limited and shared with everyone else using it.

1. Search by hand first

a. Search government documents at https://www.govinfo.gov/app/search/. Find hearings of the Senate **Committee on Foreign Relations**, sorted newest first.
b. Export the results as CSV or JSON and note the **query string** the page built for you. Part A of `govtdata01.R` downloads from an export like this.

2. How to use the GovInfo API

Documentation, with a try-it-out console: https://api.govinfo.gov/docs. Every request goes to `https://api.govinfo.gov` and carries `api_key`.

a. **Collections** — what exists, and what changed since a date: `/collections` and `/collections/CHRG/2026-01-01T00:00:00Z` (`CHRG` is Congressional Hearings; `pageSize` up to 1000; `offsetMark=*`).
b. **Search** — the website's query language, sent as JSON by `POST`:

```r
library(httr2)

body <- list(
  query      = 'collection:(CHRG) AND "Committee on Foreign Relations"',
  pageSize   = 100,
  offsetMark = "*",
  sorts      = list(list(field = "publishdate", sortOrder = "DESC"))
)

found <- request("https://api.govinfo.gov/search") |>
  req_url_query(api_key = Sys.getenv("GOVINFO_API_KEY")) |>
  req_body_json(body) |>
  req_perform() |>
  resp_body_json()
```

c. **Packages** — one document: `/packages/{packageId}/summary` returns its metadata and a `download` list (`pdfLink`, `txtLink`, `modsLink`, ...).
d. **Paging** — each response returns the next `offsetMark`; send it back to get the next page. Stop when no results come back.

3. Download the documents

a. Using Part B of `govtdata01.R`, download the **ten most recent** hearings of the Senate Committee on Foreign Relations.
b. Save them in a folder listed in `.gitignore`, and keep the download log the script writes.
c. Compare with the list you exported in section 1. Same ten documents? If not, why not?

4. Report

Write a short report on the difficulties you met.

a. How **usable** are the downloaded documents as data? What would it take to turn ten PDFs into a table or a corpus?
b. How would you **improve** the process? Consider the other formats in `download`, and what the metadata gives you without opening a file.
c. In two sentences: what did the **API** give you that the search page and last week's scraping did not?

5. Submit

a. Publish as `assign05.qmd` under **Assignments** on your website, and send the URL to the teaching assistant. Post the report and the log — **not** the PDFs, and not your key.